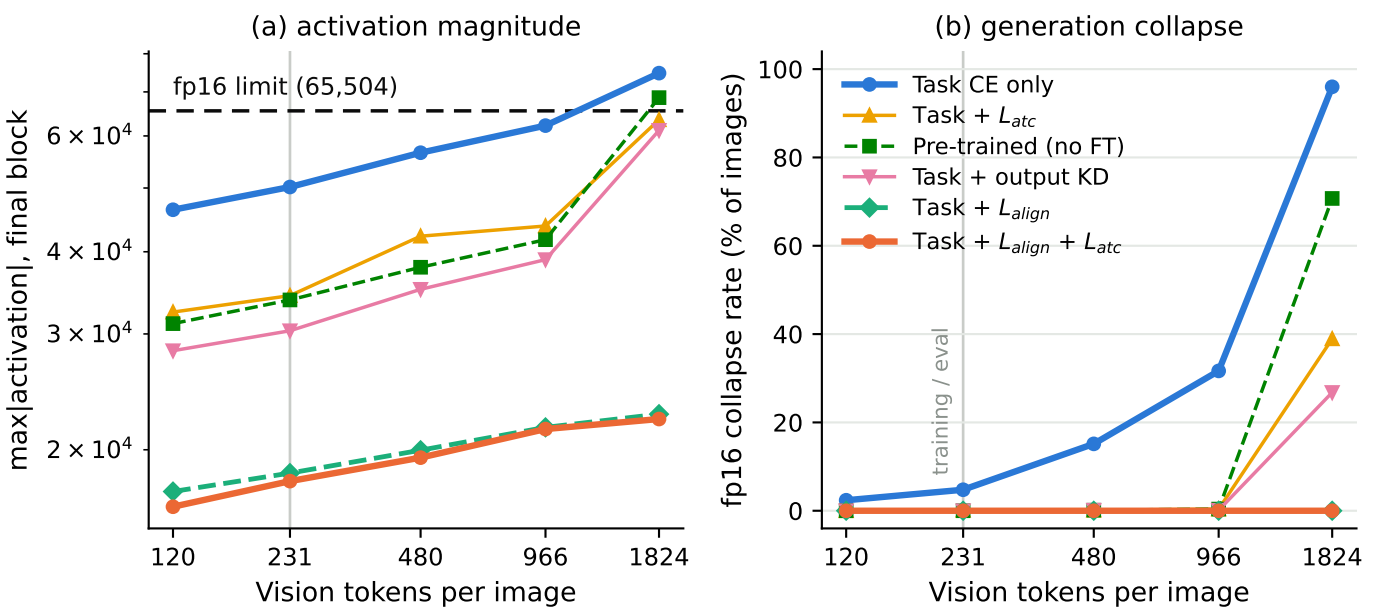
(a) activation magnitude
fp16 limit (65,504)
max|activation|, final block
6×10^4
4×10^4
3×10^4
2×10^4
120
231
480
966
1824
Vision tokens per image
(b) generation collapse
fp16 collapse rate (% of images)
100
80
60
40
20
0
Task CE only
Task + L_{atc}
Pre-trained (no FT)
Task + output KD
Task + L_{align}
Task + L_{align} + L_{atc}
training / eval
120
231
480
966
1824
Vision tokens per image